

MILL TEST CERTIFICATE



AXIS PIPE AND TUBE, LLC.
1451 Louis E. Mikulin Rd.
Bryan TX 77807

Customer: DNOW L.P. PO BOX 40985 Houston,TX,77240-0985 281-823-4700	Sales Order: 0630005623 PO: 26708945	Invoice No.: 1600052474 Date of Issue: 03/27/2025
Certificate No.: 1600052474-2		
Product Description: 5L 10.750 0.500 54.79# X52BDRH5BW-20P2		
Specification: 5L 46th Edition May 2019		

Heat Number		Lot Number		Mat Number		Steel Making / Coil Rolling																							
A416876		LT25003006		333870		Steel Dynamics Columbus, LLC , Columbus - MS - US / Steel Dynamics Columbus, LLC , Columbus - MS - US																							
Mechanical Properties																													
Tensile Test - Strip Specimen Gauge Length 2"								Temp (F) -20				CVN				Temp (F) -20													
Loc. / Orien	Size	0.5%EUL Y.S(psi)	U.T.S. Body (psi)	U.T.S. Weld (psi)	Y/T Ratio	E.L. (%)	Hardness (HRB)	Body (ft-lbf)				Shear Area (%)				Weld (ft-lbf)				Shear Area (%)				Size		Loc. / Orien	Acceptance Criteria (ft-lbf)(AVG)		
								1	2	3	Avg	1	2	3	Avg	1	2	3	Avg	1	2	3	Avg						
T180°	1 1/2"	64100	71500	79200	0.90	40.9	82	149	204	178	177	90	100	100	97	189	196	174	187	100	100	100	100	10x7.5 mm		3T90	15		
								Lateral Expasion (in)				0.086	UB	UB	0.086	Lateral Expasion (in)				UB	UB	UB	UB						

Analysis (H/P)	Chemical Composition (Max %)																	
	C	Si	Mn	P	S	Ni	Cr	Mo	Cu	V	Al	Nb	Ti	B	Ca	N	CEpcm (API Max. A.C.)	CE Results
H	0.06	0.220	0.84	0.013	0.003	0.05	0.07	0.020	0.14	0.002	0.019	0.026	0.015	0.0003	0.0019	0.0061	0.25	0.12
P	0.06	0.222	0.82	0.012	0.002	0.05	0.07	0.010	0.13	0.003	0.015	0.024	0.015	0.0002	0.0017	0.0102	0.25	0.12
P	0.06	0.223	0.83	0.011	0.002	0.05	0.07	0.010	0.13	0.003	0.015	0.024	0.015	0.0002	0.0018	0.0099	0.25	0.12

Following Controls / Tests have been satisfactorily performed:		
1.- Hydrostatic Testing:	4,860.00 psi Pressure for	5.00 seconds
2: Ultrasonic Tested to N10 - passed		
3: Seam Annealed min 1,600 F		
4: Flattening Test Passed		

Acronyms:	
L = Longitudinal	T = Transverse
Y.S. = Yield Strength	U.T.S. = Ultimate Tensile Strength
E.L. = Elongation	H = Heat Analysis
P = Product Analysis	A.C. = Acceptance Criteria
UB = Un-Broken	N/A = Not Applicable

Remarks:
1: This material meets:Grades ASTM A53/A53M-22 Grade B API 5L HFW PSL2 X52M,
2: This product meets the requirements of ASME BPVC.II.A Specification SA-53B-2023
3: Made and Melted in USA
4: Date of Manufacture 01/2025
5: This product is silicon and/or aluminum fully killed.
6: Certificate meets the requirements of ISO 10474:2013 (Inspection Certificate 3.1.B)
7: This material meets the requirements of NACE MR0175-2021 / ISO 15156:2020

We hereby certify that the material herein has been made and tested in accordance with the above specification and also with the requirement called for by the above order.
Quality Manager